

Frontend: Theory Assignment 2

1. What is CSS Box Model?

Answer: The CSS Box Model describes how every HTML element is represented as a rectangular box. It consists of four parts: **Content** – actual text/image; **Padding** – space around content; **Border** – line surrounding the padding; **Margin** – space outside the border.

Order: Content → Padding → Border → Margin

2. What is media query? Explain with a sample code.

Answer: A media query is a CSS technique used to make a website responsive according to screen size or device.

```
body {  
    background-color: white;  
}  
  
@media (max-width: 600px) {  
    body {  
        background-color: lightblue;  
    }  
}
```

3. What are the features of Bootstrap?

Answer: Bootstrap is a popular CSS framework used to create responsive and attractive websites.

- Responsive design
- Predefined CSS classes
- Grid system
- Ready-made components
- Buttons, cards, forms and navbars
- Mobile-first design
- JavaScript components
- Easy and fast development

4. Write an example of basic grid structure in Bootstrap.

Answer: Bootstrap uses a 12-column grid system.

```
<div class="container">  
    <div class="row">  
        <div class="col-4">Column 1</div>  
        <div class="col-4">Column 2</div>  
        <div class="col-4">Column 3</div>  
    </div>  
</div>
```

5. What is CSS Selectors?

Answer: CSS selectors are used to select HTML elements so that CSS styles can be applied to them.

Common selectors include element selector, class selector, ID selector, universal selector and group selector.

```
p { color: red; } /* Element */  
.box { color: blue; } /* Class */  
#heading { color: green; } /* ID */  
* { margin: 0; } /* Universal */  
h1, p { color: black; } /* Group */
```

6. What are CSS backgrounds, list the properties?

Answer: CSS background properties are used to set and control the background of an HTML element.

Important properties: **background-color**, **background-image**, **background-repeat**, **background-position**, **background-size**, **background-attachment**, **background**.

```
body {
  background-color: lightblue;
  background-image: url("image.jpg");
  background-repeat: no-repeat;
  background-position: center;
  background-size: cover;
}
```

7. How can we overlap elements in CSS?

Answer: Elements can be overlapped using positioning and the **z-index** property. A higher z-index places an element above one with a lower z-index.

```
.box1 {
  position: absolute;
  left: 50px;
  top: 50px;
  z-index: 1;
}

.box2 {
  position: absolute;
  left: 80px;
  top: 80px;
  z-index: 2;
}
```

8. What are the different positioning properties in CSS?

Answer: The five main CSS positioning values are:

Static: default position.

Relative: moves relative to its normal position.

Absolute: positioned relative to its nearest positioned ancestor.

Fixed: fixed relative to the browser window.

Sticky: acts like relative until a scroll position, then behaves like fixed.

9. What is ECMAScript?

Answer: ECMAScript is a standardized scripting-language specification on which JavaScript is based. It defines features and rules for JavaScript, such as variables, functions, classes, objects, modules and promises. ES6 (ECMAScript 2015) introduced features such as let, const, arrow functions and classes.

10. Difference between var, let and const. Explain Temporal Dead Zone.

Answer: var: function-scoped, can be redeclared and reassigned.

let: block-scoped, cannot be redeclared in the same scope, but can be reassigned.

const: block-scoped, cannot be redeclared or reassigned.

Temporal Dead Zone (TDZ): the period between entering a block and the initialization of a let or const variable. Accessing it before initialization causes a ReferenceError.

```
var a = 10;
a = 20;      // Allowed

let b = 10;
b = 20;      // Allowed

const c = 10;
c = 20;      // Error

console.log(x); // ReferenceError
let x = 10;
```